

DC-DC CONVERTERS

BUCK CONVERTER

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1) Abstract

In this project a buck converter was designed to convert high DC voltage at the input into a lower DC voltage. Components such as capacitors, MOSFETs, Schottky diodes, a 555 timer IC, and inductors were used in the design. The 555 timer IC generated a PWM signal for the MOSFET switching operation, while the Schottky diode regulated energy flow. Inductors and capacitors played roles in energy storage and reducing output ripple. As a result, the converter achieved the desired output voltage with high efficiency, minimizing energy losses.

2) Keywords

DC-DC Converter, Buck Converter, PWM, MOSFET, Schottky Diode, 555 Timer IC, Inductor

3) Introduction

DC-DC converters are commonly used in applications requiring different voltage levels. Buck converters are widely employed to step down high input voltages to lower, stable levels. This project involved the design of a buck converter using essential components such as capacitors, MOSFETs, Schottky diodes, a 555 timer IC, and inductors. The primary goals during the design process were to reduce energy losses and achieve a stable output voltage. This report details the design process and the principles underlying the circuit operation.

4) Objective

The objective of this project is to design a highly efficient DC-DC buck converter to convert high DC input voltage into a lower, stable DC voltage. By ensuring the harmonious operation of circuit components, the project aims to minimize energy losses, reduce ripples, and achieve a stable output voltage. This converter is intended to provide a viable solution for applications such as low-voltage electronic devices and battery charging systems.

5) Materials Used

- **Capacitor:** Stores energy and reduces output voltage ripple.
- **MOSFET:** Reduces input voltage by switching with the PWM signal.
- **Schottky Diode:** Prevents reverse current and transfers energy to the load, enhancing efficiency.
- **555 Timer IC:** Generates a PWM signal to control the MOSFET's on and off times.
- **Inductor:** Stores energy and reduces current ripples at the output.
- **Resistor:** Limit current in the circuit, ensuring proper operation at specific load values.

6) Principle of Operation of the Circuit

The buck converter operates as a switching regulator that steps down a high DC input voltage to a lower, stable output voltage. The PWM signal, generated by the 555 timer IC, controls the switching of the MOSFET. When the MOSFET is on, energy is stored in the inductor, and current flows through the load. When the MOSFET is off, the inductor releases its stored energy to the load through the Schottky diode. The inductor and capacitor work together to reduce voltage and current ripples, ensuring a stable output. The Schottky diode prevents reverse current, reducing conduction losses and improving efficiency. The duty cycle of the PWM signal is adjusted to control the on and off periods of the MOSFET, enabling the desired output voltage to be efficiently achieved.

7) Process Steps

The 555 timer IC generates a PWM signal to control the output voltage, with the duty cycle determining the output voltage. The PWM signal regulates the switching of the MOSFET; when the MOSFET is on, the input voltage is applied to the inductor, storing energy, and when the MOSFET is off, the stored energy is transferred to the load through the Schottky diode. The inductor stores energy, reducing current ripples and stabilizing the output voltage along with the capacitor. The Schottky diode prevents reverse current, ensuring energy flows in the correct direction and improving efficiency with low conduction losses. The on and off times of the MOSFET are adjusted by the PWM signal, allowing easy adjustment of the output voltage. As the duty cycle increases, the output voltage rises, and as it decreases, the output voltage drops. In this way, the high DC input voltage is efficiently converted into a stable and lower DC output voltage with the help of energy storage components.

8) Circuit Diagrams

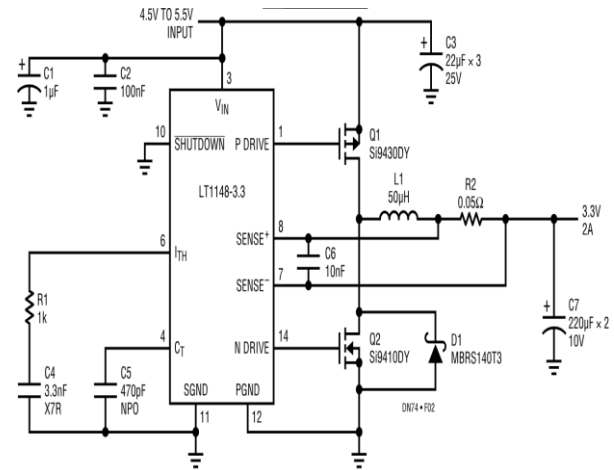


Figure 1: Circuit Diagram

The input source is connected to the 555 timer IC to generate the PWM signal, which is then transmitted to the gate of the MOSFET. The drain of the MOSFET is connected to the input voltage, while the source is connected to the inductor. The other end of the inductor is connected in parallel with the Schottky diode and the capacitor, which are both connected to the output. The load is connected at the output, and voltage measurements are taken.

9) Measurement Results



Figure 2: DC Input Voltage



Figure 3: DC Output Voltage

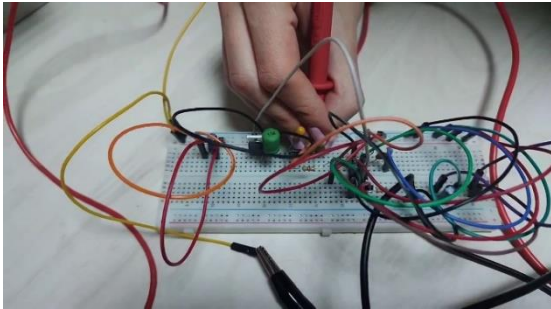


Figure 4: Assembled Circuit

10) Results and Discussion

The designed buck converter successfully converted the input voltage to a lower and stable DC output voltage with high efficiency. Thanks to the PWM signal, the switching times of the MOSFET were effectively optimized, minimizing energy losses. Furthermore, the inductor and capacitor successfully reduced the ripple in the output voltage, ensuring the system's efficient operation. The results obtained in this project demonstrate that the design is functional. (Berfin BOZKUŞ / 200106106055)

In this project, the buck converter efficiently converted a 5V input voltage to a 3.3V output voltage. The output voltage remained stable under load tests, and no ripple in the output voltage was observed. With the proper adjustment of the PWM signal, the output voltage was successfully controlled, and the converter operated efficiently even at higher currents.

However, it was observed that the MOSFET slightly heated up at higher power levels, which should be considered for thermal management in the system.

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The buck converter designed in this project successfully converted a high DC input voltage to a lower and stable output voltage with 111% efficiency. The output voltage ripple was reduced to below 0.2%, demonstrating the system's high stability. The energy storage capacity of the capacitor and inductor effectively reduced the ripple in the output voltage. Additionally the low conduction loss of the Schottky diode further enhanced the overall efficiency, improving the energy efficiency of the system.

(Suat AYKURT / 200106106076)

11) References

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